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**Engineering Support for Design, Development, and Integration of
RF and HF Systems for the Advanced Radar Systems Branch**

Statement of Work (SOW)

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1.0. Introduction

The Naval Research Laboratory (NRL) Radar Division is seeking contractor assistance in the design, development, and integration of radio frequency (RF) and high frequency (HF) systems. The task areas of interest include: mobile over-the-horizon (OTH) radar, HF surface wave radar, arctic HF OTH radar, homeland defense radar, high power HF transmitter, and multifunctional RF.

1.1 Scope

This statement of work (SOW) describes the research and development, engineering, and integration support the contractor is required to provide the Advance Radar Systems Branch of the Radar Division at NRL, Washington DC. NRL Radar Division requires the contractor to work alongside the Government research team in the following areas: HF and RF systems, signal processing hardware, mechanical analysis, and system integration on mobile platforms. Additionally, this SOW requires the contractor to support technology advancements, system requirements definition, architecture maturation, system and unit design, and the development and transition to an operational system to support the effort. The contractor will be required to support unclassified and classified tasks up to the Secret level.

2.0. General Requirements

2.1 Non-Personal Services

The Government will neither supervise contractor employees nor control the method by which the contractor performs the required tasks. The Government will not assign tasks to, nor prepare work schedules for individual contractor employees. The contractor shall be responsible for managing its employees and guarding against any actions that are of the nature of personal services, or give the perception of personal services as defined in FAR 37.104, Personal Services Contracts.

The contractor shall notify the Contracting Officer if any Government requested actions constitute, or are perceived to constitute, personal services.

2.2 Overview

The following subsections specify requirements for contract management and contractor personnel administration.

2.2.1 Management

The contractor shall establish clear organizational lines of authority and responsibility to ensure effective management of the resources assigned to this contract.

2.2.2 Personnel Administration

The contractor shall assign a single point of contact, also known the Program Manager (PM), who must work closely with the Contracting Officer and the Contractor Officer's Representative (COR), as applicable.

The contractor PM is ultimately responsible for ensuring that the contractor's performance meets all Government contracting requirements within cost and schedule. The PM must have the requisite authority for full control over all company resources necessary for contract performance. As appropriate, the role of PM is a full-time management position.

The contractor shall maintain the adequacy of its employees by providing initial and refresher training to meet the SOW requirements.

The contractor shall make all necessary travel arrangements for contractor employees only after the travel has been reviewed and approved by the COR at least three weeks prior to the actual trip. Exceptions may be made for unexpected travel requirements.

2.2.3 Administration

The contractor shall establish processes and assign appropriate resources to effectively communicate with the Government during contract administration.

The contractor shall respond to Government requests for contractual actions within five business days.

The contractor shall assign work effort and maintain proper and accurate time keeping records of personnel assigned to work on this contract.

2.2.4 Subcontractor Management

The contractor shall be responsible for any subcontract management necessary to integrate work performed on this contract and shall be responsible and accountable for subcontractor performance on this contract. The prime contractor will manage work distribution to ensure there are no Organizational Conflict of Interest (OCI). Consent to subcontract shall be in accordance with procedures established under FAR clause 52.244-2, Subcontracts.

2.3 Contractor Furnished Equipment, Materials, Subcontracts and Supplies

Equipment and unexpended materials and supplies purchased by the contractor under this contract become the property of the Government at the end of the performance period, including all options.

The contractor shall provide any other equipment, material, and supplies, not furnished by the Government, but required to perform the work detailed herein.

2.4 Contractor Personnel, Disciplines, and Specialties

The required and/or desired minimum education, certifications, clearance, and experience requirements for contractor personnel to perform the support tasks identified in this SOW are defined in the contracts "Personnel Qualifications" attachment.

2.5 Use of Government Property at Government Facilities

If determined to be in the Government's best interest, the Government will make available the materials, office space, communications capability, and information for use by the contractor in

the performance of this contract at Government facilities. Government property used by the contractor at the Government's facilities will be on a rent-free basis and all such property shall be considered to remain in the possession and control of the NRL for property responsibility and accountability purposes.

2.6 Requirement for On-site Contractors

The contractor shall comply with requirements for On-Site Contractor performance at Government facilities. The full text of the requirements for On-Site contractors is set forth as an attachment to the contract. Pursuant to DFARS 211.106 contractor employees shall identify themselves as contractor personnel by introducing themselves or being introduced as contractor personnel and displaying distinguishing badges or other visible identification for meetings with Government personnel. In addition, contractor personnel shall appropriately identify themselves as contractor employees in telephone conversations and in formal and informal written correspondence.

2.7 Place of Performance

It is anticipated that work will be performed at the following locations:

- NRL – Washington, D.C.
- NRL Chesapeake Bay Detachment (CBD) – Chesapeake, MD
- Tilghman Island, MD
- Naval Air Station Patuxent River – Patuxent River, MD
- NASA Flight Facility – Wallops Island, VA
- Other Government facilities as needed
- Contractor facilities

The contractor shall perform work at Government facilities within the normal work hours unless differing hours are specified by the COR. The contractor is not required to maintain the same hours as Government employees; however, contractor employees performing working at NRL must work during the normal workweek. Contractor personnel working off-site must remain reachable (e.g. phone, email, Microsoft Teams, etc.) by Government personnel. In the event of severe weather conditions, the contractor should refer to the top banner of the NRL website for the NRL-DC operating status or refer to the Office of Personnel Management website for the operating status of the U.S. Government offices in Washington, D.C. area and for additional information.

2.8 Travel / Temporary Duty (TDY)

Travel may be required to conduct experimental research, and/or attend Government reviews or scientific meetings and seminars. The contractor shall submit all travel requirements (e.g. plans, agenda, itinerary, and dates) in writing for pre-approval to the Government COR. Travel is on a strictly cost-reimbursable basis. The contractor shall bill costs for travel in accordance with FAR 31.205-46 Travel Costs (subject to local policy & procedures) and DoD's Joint Travel Regulations.

The contractor shall be able to provide two persons within one day of travel notification for deployment to support work described in this SOW at any location within the United States. The persons are to deploy with tooling for mechanical or electrical work necessary to maintain or repair EMWRF related equipment.

2.9 Patents – Reporting of Subject Inventions

All subject inventions must be properly reported through the iEdison system at <https://www.nist.gov/iedison>. Additionally, the contractor shall provide interim reports and a final report in accordance with DFARS 252.227-7039 Patents - Reporting of Subject Inventions, utilizing DD Form 882. If there are no subject inventions to report, the contractor is still required to submit the DD Form 882. Reports shall be delivered via email to the points of contact identified in Block 14 of CDRL A006.

3.0. Technical Requirements

3.1 Mobile Over-the-Horizon Radar R&D (Task 1)

The objective of this task is to develop, design, install, and maintain a long-range surveillance and targeting system to demonstrate and mature HF radar and communications on mobile platforms. Under this contract, the contractor shall provide support in the following areas:

- High Frequency (HF) radar system design (CDRL A002) (CDRL A005)
 - Perform system analysis, design, development, and laboratory and/or field test of HF systems (i.e. radar, communications, electronic warfare)
 - Conduct trade studies that lead to optimal system performance for a defined set of conditions and applications
 - Perform technical analysis and computer modeling and simulations to evaluate component and subsystem level performance for the radar system
- Design and development of signal processing radar hardware and software (CDRL A002) (CDRL A005)
 - Develop, design, and build new signal processing hardware for the system
 - Perform laboratory and field experiments of radar sub-systems to support RF development
 - Setup and perform component and system measurements such as S-parameters, linearity (i.e. P1dB, OIP3) noise figure and others
 - Develop software blocks to integrate and mature various signal processing blocks (i.e. beamforming, peak detection, tracking, and coordinate registration)
- Carry out system integration of transmit and receive HF systems on land, maritime, and/or air platforms (CDRL A002)
 - Develop a schedule to plan out system integration and installation timeline
 - Develop a procurement plan in support of equipment acquisition
 - Develop test plans, reports, user manuals, and other documentation
- Participate in technical meetings and design reviews, provide critiques, review documentation, and monitor technical progress (CDRL A003)
- Perform mechanical analysis for installation of antenna systems and other experimental systems including (CDRL A002)
 - Thermal analysis

- Wind loading and vibration analysis
- Development of antenna laydown plans for HF systems
- Development of CAD models

3.2 HF Surface Wave Radar R&D (Task 2)

The objective of this effort is to design, develop and install a wideband surface wave system to demonstrate long-range detection of surface targets. Under this task, the contractor shall perform the following tasks:

- Design, prototype, and install wideband HF surface antenna systems (CDRL A002)
 - Develop mature mechanical designs of antenna systems capable of surviving harsh beach environments and built for long periods
 - Carry out wind loading and vibration analysis
 - Carry out install of the antenna systems with limited oversight OCONUS
- Carry out system integration of transmit and receive HF systems on land (CDRL A002)
 - Develop a schedule to plan out system integration and installation timeline
 - Develop a procurement plan in support of equipment acquisition
 - Develop test plans, reports, user manuals, and other documentation
 - Oversee/work with subcontractors during the build phase of the system
- Participate in technical meetings and design reviews, provide critiques, review documentation, and monitor technical progress (CDRL A003)
- Perform data analysis and processing of data collected from data collection exercises (CDRL A002)

3.3 Arctic HF OTH Radar R&D (Task 3)

The objective of this effort is to design, develop, and install a wideband multi-function receive systems to demonstrate long-range detection of low altitude targets in the artic/polar environment. Under this task, the Contractor shall perform the following tasks:

- Design, prototype, and install wideband HF multi-function receive systems (CDRL A002)
 - Develop mature mechanical designs of antenna systems capable of surviving harsh artic/polar environments and built for long periods
 - Carry out wind/ice/snow loading and vibration analysis
 - Develop and demonstrate array calibration techniques
 - Carry out installation of the antenna systems with limited oversight OCONUS
- Carry out system integration of transmit and receive HF systems on land (CDRL A002)
 - Develop a schedule to plan out system integration and installation timeline
 - Develop a procurement plan in support of equipment acquisition
 - Develop test plans, reports, user manuals, and other documentation
 - Oversee/work with subcontractors during the build phase of the system
- Participate in technical meetings and design reviews, provide critiques, review documentation, and monitor technical progress (CDRL A003)
- Perform data analysis and processing of data collected from data collection exercises
 - Develop signal processing techniques to mitigate auroral effects
 - Develop signal processing techniques for range-Doppler processing of noise waveforms

3.4 Homeland Defense Radar R&D (Task 4)

The objective of this effort is to design, develop, and install an HF over-the-horizon skywave radar for the detection and tracking of aircraft. Under this task, the contractor shall perform the following tasks:

- Conduct performance analysis of over-the-horizon radar configurations using relevant radar system, environmental, and target models. (CDRL A002)
- Develop advanced tracking algorithms (CDRL A002) (CDRL A005)
 - Automated tracking
 - Multi-mode track assignment
- Develop advanced arctic ionospheric models (CDRL A002) (CDRL A005)
- Develop new signal processing suited for 2D receive arrays (CDRL A002) (CDRL A005)
- Conduct physical site planning and environmental analysis (CDRL A002)
- Develop new background noise models based on collected site noise data (CDRL A002) (CDRL A005)
- Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress (CDRL A003)

3.5 High Power HF Transmitted R&D (Task 5)

The objective of this task is to design, build, and install a modular transmit system on a mobile platform (e.g. barge, Offshore Supply Vessel (OSV)), or on land. At the end of the effort, the Government intends to demonstrate the ability to design a system that can be setup and disassembled in a short time such as a 2-3 weeks and can be modularly built up to any size that is required. Under this task, the contractor shall perform the following tasks:

- Design, prototype and install wideband HF surface and/or skywave antenna systems (CDRL A002) (CDRL A005)
 - Develop mature mechanical designs of antenna that can be installed on any platform or land area. These antennas should be capable of surviving harsh environments and should be installed and de-installed easily
 - Carry out all force analysis for a mature design
 - Carry out installation of the antenna systems with limited oversight
- Carry out a full system integration of the transmit array on platform/area as needed (CDRL A002) (CDRL A005)
 - Develop a detailed schedule to plan out system integration and installation timeline
 - Develop a procurement plan in support of equipment acquisition
 - Develop test plans, reports, user manuals and other relevant documentation
 - Oversee/work with sub-contractors tasking during design/build phase of the program
- Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress (CDRL A003)
- Support NRL engineers during all data collection events (CDRL A002)

3.6 HF Surface Wave Radar R&D (Task 6)

The objective of this effort is to design, develop, and install multifunction RF (e.g. radar, communications, electronic warfare) systems. Under this task, the contractor shall perform the following tasks:

- Design, prototype, and install multifunction RF antenna systems (CDRL A002) (CDRL A005)
 - Develop mature mechanical designs of antenna systems and ancillary equipment capable of surviving harsh environments
 - Carry out structural, vibrational, and thermal analysis using modeling tools (e.g. – Solid Works, Ansys, etc.)
- Carry out system integration of multifunction RF systems on land, maritime, or airborne platforms (CDRL A002) (CDRL A005)
 - Develop a schedule to plan out system integration and installation timeline
 - Develop a procurement plan in support of equipment acquisition
 - Develop test plans, reports, user manuals and other documentation
 - Oversee/work with sub-contractors during the build phase of the system
- Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress (CDRL A003)
- Perform data analysis and processing of data collected from data collection exercises (CDRL A002)

3.7 Technical Program Support

The contractor shall provide general, administrative, and technical support to the Government on multiple programs spanning multiple sponsors by reporting monthly on financial expenditures (CDRL A001) and technical progress (CDRL A002). Additionally, the contractor shall provide documentation of technical information provided, and decisions and agreements reached, at meetings and conferences (CDRL A003).

4.0. Deliverables

Data Item No.	Title of Data Item
A001	Funds and Labor Hour Expenditure Report
A002	Scientific and Technical Reports
A003	Meeting Minutes
A004	Government Property (GP) Inventory Report
A005	Software/Hardware Design Description (SHDD)
A006	Patents - Reporting of Subject Inventions

5.0. Special Requirements

This section describes the special requirements for this effort. The following subsections provide details of various considerations for this effort.

5.1 Security

Collateral SECRET clearances will be required for technical personnel. Interim-SECRET clearances are allowed on a temporary basis for new-hires, given the below considerations.

Contractor personnel assigned to this contract, who require access to Secret information, must have a favorably completed Tier 3 investigation and a final Secret clearance at time of proposal submission. NRL base access requires a favorably completed Tier 3 investigation and a DoD granted final Secret clearance.

Contractors issued a DoD interim Secret security clearance may ONLY work on unclassified tasks as outlined in the SOW. Unclassified IA or IT positions are NOT authorized with a DoD issued interim Secret clearance.

Contract Facility Security Officers (FSOs), and/or NRL CORs, are required to report immediately to Code 1230 any derogatory information regarding a contractor holding an interim-SECRET clearance, either on-site or off-site. FSOs and NRL CORs shall also notify Code 1230 when a contractor's interim or final clearance has been declined or revoked. Individual installation access restrictions may apply.

For contractors working on unclassified task orders and performing at the locations listed in block 8 of the DD 254, a favorably completed suitability determination and a submitted T3 investigation is required prior to the start of work. Individual installation access restrictions may apply. All contractors (including subcontractors) shall supplement their current security practices by requiring any personnel involved in executing the contract to complete Operations Security (OPSEC) training via DCSA through their CDSE site: <https://securityawareness.usalearning.gov/opsec/index.htm>; and follow any OPSEC guidance that may pertain to the project.

Contractors with an Interim SECRET clearance must be escorted at all times in classified spaces. Contractors with an Interim SECRET clearances are prohibited from having any access to the following types of classified information:

- Communications Security (COMSEC)
- Special Compartmented Information (SCI)
- Special Access Program (SAP) Information
- North Atlantic Treaty Organization (NATO) Information
- Critical Nuclear Weapons Design Information (CNWDI)
- Restricted Data (RD) (and Formerly Restricted Data)
- Naval-Nuclear Propulsion Information (NNPI)
- Foreign Government Information (FGI)

5.2 Safety

The contractor shall comply with the NRL Occupational Safety and Health Manual 5100.13C, the Safety Related Work Practices NFPA 70E-2021 and the NRL Security Manual, 5510.40D with respect to all contractor employees working at NRL.

5.3 Government Furnished Property

The Government may provide Government property to the contractor for use in the performance of this contract, which may include materials, equipment, laboratory spaces, and facilities as needed for contractor personnel working off-site of NRL.

Inventories of GFP shall be reported via CDRL A004 as applicable, and maintained in the GFP Module located in the Procurement Integrated Enterprise Environment (PIEE). No other material, labor, or facilities will be furnished by the Government unless approved by the COR or Contracting Officer. If GFP/GFE/GFI is required for the performance of this effort, the contract shall clearly identify what is required and the time it is necessary for uninterrupted performance of the effort.

5.4 Travel

The contractor shall be required to travel periodically in support of the efforts described in this SOW. Travel requirements include day trips to other Government locations within the local Washington, D.C. commuting area and other CONUS travel.

(End of SOW)